Part – C:

Metric	**Value**
Worst slack at T = 10 ns	7.40 ns
Tmin	2.61 ns
Fmax	1000 / 2.61 = **383.1 MHz**
Slack at T = Tmin	**+0.006 ns**
Critical path start point	_095_ = a_r[1] (input register)
Critical path end point	_092_ = p[6] (output register)
Numbers of cells on the critical path	8 combinational (10 with the two flip-flops)
Latency (2 × Tmin)	5.22 ns
Throughput	383.1 million results/s

```
/foss/designs/mult4_rtl2gds > T=10 sta sta/sta.tcl
OpenSTA 3.1.0 f89887b596 Copyright (c) 2026, Parallax Software, Inc.
License GPLv3: GNU GPL version 3 <http://gnu.org/licenses/gpl.html>

This is free software, and you are free to change and redistribute it
under certain conditions; type `show_copying' for details.
This program comes with ABSOLUTELY NO WARRANTY; for details type `show_warranty'.
worst slack max 7.40
/foss/designs/mult4_rtl2gds > T=2.61 sta sta/sta.tcl
OpenSTA 3.1.0 f89887b596 Copyright (c) 2026, Parallax Software, Inc.
License GPLv3: GNU GPL version 3 <http://gnu.org/licenses/gpl.html>

This is free software, and you are free to change and redistribute it
under certain conditions; type `show_copying' for details.
This program comes with ABSOLUTELY NO WARRANTY; for details type `show_warranty'.
worst slack max 0.01
/foss/designs/mult4_rtl2gds >
```

list of cells on the critical path

#	Instance	Cell	Delay (ns)	Arrival (ns)
start	_095_	dfrtp_1 (clk to Q)	0.398	0.398
1	_039_	clkinv_1	0.074	0.472
2	_047_	o311ai_0	0.348	0.820
3	_050_	a21oi_1	0.172	0.992
4	_063_	maj3_1	0.381	1.373
5	_064_	xnor2_1	0.160	1.534
6	_066_	maj3_1	0.385	1.918
7	_075_	maj3_1	0.377	2.295
8	_079_	xnor3_1	0.175	2.471
end	_092_	dfrtp_1 (setup 0.134)		